

# Route decision: Mumbai JNPT Logistics Park → Chennai Logistics Park

HUB-MUMBAI → HUB-CHENNAI · mode any (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC

**Recommendation: SAFEST / BALANCED**, ROAD via direct. Expected arrival in **31 h** (1.3 days); plan for 33 h (p85), worst case 35 h (p95). Cost: freight AED 6,937. Peak leg risk 0%.

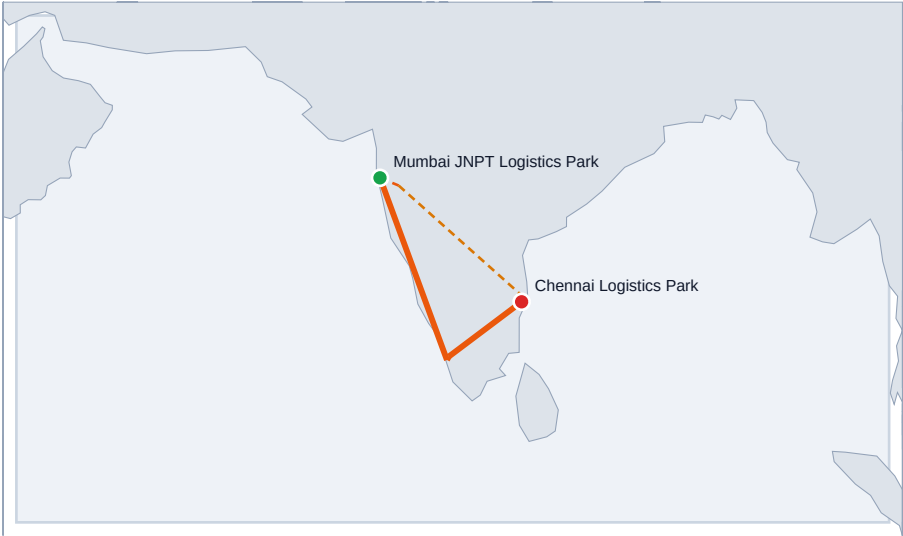
## Options

| Option            | Mode | Via    | Plan h | p50 h | p85 h | p95 h | Freight    | Risk |
|-------------------|------|--------|--------|-------|-------|-------|------------|------|
| FASTEST           | AIR  | direct | 15     | 25    | 29    | 32    | AED 51,951 | 2%   |
| SAFEST / BALANCED | ROAD | direct | 23     | 31    | 33    | 35    | AED 6,937  | 0%   |

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



## Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 25.0 h (p85 29.4 h, p95 32.2 h), AED 51,951 freight, 2 handoffs. 5.9 h faster than the next option. AED 45,014 more (freight) than SAFEST.

| # | Mode    | To                            | km  | Hours | Delay p85 | Cost       | Risk | Signal |
|---|---------|-------------------------------|-----|-------|-----------|------------|------|--------|
| 1 | ROAD    | Pune Air Cargo Hub            | 106 | 1.6   | +3.6      | AED 465.71 | 0%   |        |
| 2 | handoff | Pune Air Cargo Hub            |     | 6.0   | +0.0      | AED 550.88 | 2%   |        |
| 3 | AIR     | Chennai International Airport | 912 | 1.1   | +8.8      | AED 50,268 | 0%   |        |
| 4 | handoff | Chennai International Airport |     | 6.0   | +0.0      | AED 550.88 | 2%   |        |
| 5 | ROAD    | Chennai Logistics Park        | 26  | 0.4   | +3.6      | AED 115.32 | 0%   |        |

Transit 3.1 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 15.1 h. Model delay buffer p50/p85/p95: +10.0 / +16.1 / +20.7 h. Route band p50 25.0 h, p85 29.4 h, p95 32.2 h (independent legs); 35.7 h if all legs are late together. Costs: linehaul AED 50,849, transfers AED 1,102, disruption surcharge AED 0.00.

Why the slowest leg (AIR-PUNE → AIR-CHENNAI, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST / BALANCED: ROAD via direct

ROAD via no chokepoints; expected 30.9 h (p85 32.8 h, p95 34.6 h), AED 6,937 freight, 0 handoffs. 5.9 h slower than FASTEST. AED 45,014 cheaper (freight) than the lowest-cost alternative.

| # | Mode | To                                  | km    | Hours | Delay p85 | Cost      | Risk | Signal |
|---|------|-------------------------------------|-------|-------|-----------|-----------|------|--------|
| 1 | ROAD | Kochi Logistics Distribution Center | 1,060 | 15.6  | +3.6      | AED 4,670 | 0%   |        |
| 2 | ROAD | Cochin International Airport        | 22    | 0.3   | +3.6      | AED 95.26 | 0%   |        |
| 3 | ROAD | Chennai Logistics Park              | 493   | 7.2   | +3.6      | AED 2,172 | 0%   |        |

Transit 23.1 h + transfers 0.0 h + announced scenario delay 0.0 h = plan 23.1 h. Model delay buffer p50/p85/p95: +7.8 / +10.9 / +14.1 h. Route band p50 30.9 h, p85 32.8 h, p95 34.6 h (independent legs); 37.2 h if all legs are late together. Costs: linehaul AED 6,937, transfers AED 0.00, disruption surcharge AED 0.00.

Why the slowest leg (HUB-MUMBAI → DC-KOCHI, p85 3.6 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.